

2. Dataset Overview: Medical Cost Personal Datasets

The `insurance.csv` dataset contains individual medical records:

- `age` : Age of primary beneficiary (integer)
- `sex` : Insurance contractor gender (`female` , `male`)
- `bmi` : Body mass index (kg/m^2) (ideal: 18.5 to 24.9)
- `children` : Number of children / dependents covered
- `smoker` : Smoking status (`yes` , `no`)
- `region` : Beneficiary's residential area in US (`northeast` , `northwest` , `southeast` , `southwest`)
- **Target** `charges` : Individual medical costs billed by health insurance (\$).

3. Categorical Encoding (Dummy Variables)

Linear models cannot process string variables directly. We apply **One-Hot Encoding** (or dummy encoding with `drop_first=True` to prevent the dummy variable trap of multi-collinearity):

$\text{smoker_yes} = \begin{cases} 1 & \text{if smoker} = \text{'yes'} \\ 0 & \text{if smoker} = \text{'no'} \end{cases}$

When a categorical feature has k levels, we generate $k - 1$ binary indicators. The omitted level serves as the reference baseline absorbed into the intercept b .

4. Model Training & Interpreting Learned Coefficients

In scikit-learn:

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
```

The prediction equation is:

$$\hat{y} = b + w_1x_1 + w_2x_2 + \cdots + w_px_p$$

- **Intercept b :** The expected baseline charges when all numerical features are zero and categorical indicators are at reference values.
- **Coefficient w_j :** The expected change in target charges (Δy) for a 1-unit increase in feature x_j , holding all other features strictly constant (*ceteris paribus*).
 - For example, smoking status yields an enormous positive coefficient $w_{\text{smoker_yes}} \approx +23,800\$$, reflecting massive risk premiums.